

Luke Peter

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EDUCATION

University of South Carolina, Columbia, SC

Aug 2022 - May 2026

Bachelor of Science in Computer Engineering, Minor in Electrical Engineering, GPA: 3.919

Relevant course work: Embedded Systems, Capstone Comp. Engr. Project, Advanced Digital Design, Artificial Intelligence, Computer Networks, Building Secure Software, Operating Systems

Honors/Awards: President's List, Capstone Scholar, Palmetto Fellow, Dean's Scholar

PROJECTS & EXTRACURRICULAR

Embedded Systems Capstone Project | ESP32-S3, C, Hardware Integration

Spring 2026

- Designed and implemented multiple embedded systems using C on ESP32-S3, integrating peripherals including DC motors, LEDs, audio output (speaker), and power systems.
- Developed low-level communication protocols, including bit-banged I2C to interface with MPU-6050 sensors to be used as gyroscope.
- Debugged and validated system behavior using oscilloscopes and signal analysis to troubleshoot timing and hardware communication issues.
- Integrated hardware and software components to build responsive, real-time PID control and sensing.

Robotics Survey Paper

April 2026

- Analyzed and synthesized research on Deep Learning in Robot Perception. Developed a strong understanding of tradeoffs of traditional machine learning vs deep learning methods, as well as robot vision techniques.

University of South Carolina Club Basketball

August 2024 - May 2026

- Selected as team captain for competitive NCBBA club basketball team contributing to team leadership

PROFESSIONAL EXPERIENCE

South Carolina Department of Commerce, Columbia, SC

July 2024 - Present

Information Technology Intern

- Assisted in migrating the agency's on-premises server infrastructure to Microsoft Azure, improving system scalability and reliability, and becoming the first SC state agency to become fully cloud-based.
- Provided tech support and education for over 100 employees, resolving hardware, software, and network issues

Waiter's Choice Catering, Charlotte, NC

May 2023 - Jan 2024

Caterer

- Devised creative solutions in a high-pressure environment, and trained recruits for events with over 200 guests

SKILLS

Programming languages: Java, C, C++, Python, RISC-V Assembly, Verilog, SQL

Computer software/ frameworks: VS Code, Eclipse, Arduino IDE, Linux command line, Quartus, ModelSim, MATLAB, LT Spice, MySQL, Microsoft Office Suite